

Genetic architecture of complex traits in plants

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Genetic architecture refers to the numbers and genome locations of genes that affect a trait, the magnitude of their effects, and the relative contributions of additive, dominant, and epistatic gene effects. Quantitative trait locus (QTL) mapping techniques are commonly used to investigate genetic architectures, but the scope of inferences drawn from QTL studies are often restricted by the limitations of the experimental designs. Recent advances in experimental and statistical procedures, including the simultaneous analysis of QTL that segregate in diverse germplasm, should improve genetic architecture studies. High-resolution QTL mapping methods are being developed that may define the specific DNA sequence variants underlying QTL. Studies of genetic architecture, combined with improved knowledge of the structure of plant populations, will impact our understanding of plant evolution and the design of crop improvement strategies.

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Introduction

Many traits that are important for fitness and agricultural value of plants are complex quantitative traits, affected by many genes, the environment, and interactions between genes and environments. QTL mapping is a key tool for studying the genetic architecture of complex traits in plants, facilitating estimation of the minimum number of genome regions that affect a trait, the distribution of gene effects, and the relative importance of additive and non-additive gene action [1,2]. Here, I review difficulties of QTL mapping, and focus on recent advances in both experimental techniques and statistical analyses aimed at overcoming these difficulties and providing robust estimates of the key parameters of genetic architecture in plants. Genetic architecture is defined by knowledge of the genes controlling a trait, and QTL of moderate and even small effects are now being resolved to the gene

level [3,4]. Concomitantly, new QTL approaches explicitly incorporate the genetic heterogeneity of diverse gene pools to obtain more robust and comprehensive understanding of genetic architecture. Finally, novel approaches are being pursued to better understand epistasis and heterosis in plants.

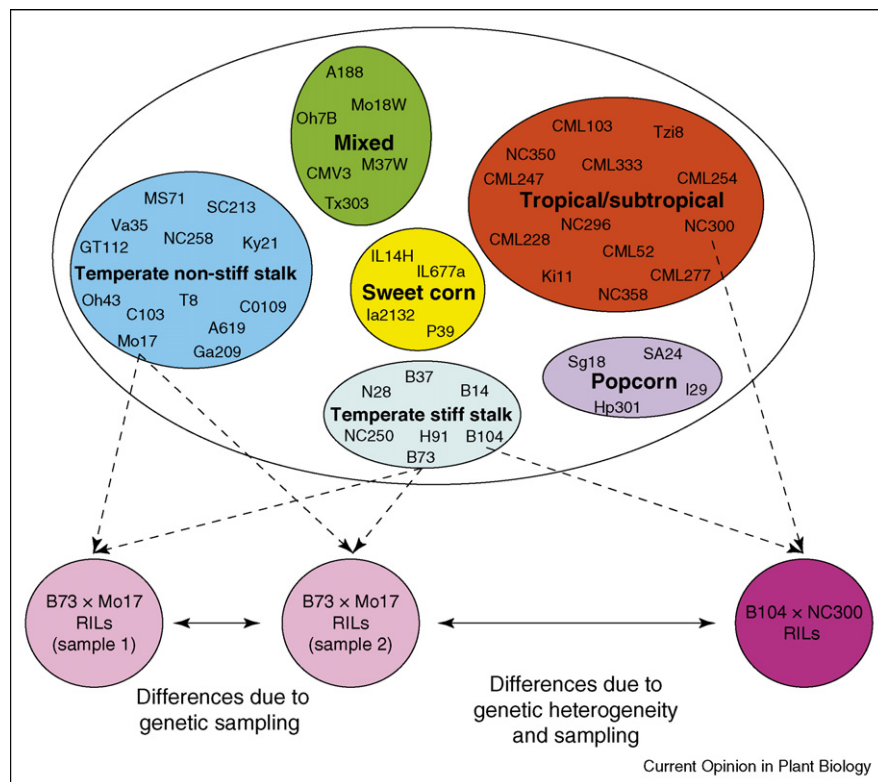
Genetic architecture

Most plant QTL studies have been based on individual biparental populations. Inferences from these studies suggested that plant populations segregate for a limited set of small-effect QTL plus a very few QTL that have large effects [3]. Further, QTL were often not consistently detected across mapping populations [5]. This is due, in part, to genetic heterogeneity – when many genes control a trait, different subsets can segregate in different populations (Figure 1). In addition, sampling limited numbers of progeny in mapping studies tends to cause exactly the skewed distribution of QTL effects and the limited number of QTL observed in many studies, even if many genes with equal and small effects actually control the trait (Figure 1; [5]).

Because QTL mapping in small populations confounds the effects of QTL with statistical artifacts caused by sampling, researchers have begun to study complex trait genetics in large populations. Recent analyses of very large maize populations revealed a substantially more complex genetic architecture than was previously suggested in QTL mapping studies [2,6]. Relatively large numbers of QTL were detected for both seed oil content [2] and grain yield [6], and in both studies, the detected QTL explained only half or less of the total genetic variation. The absence of QTL with large effects in these studies does not necessarily imply that large effect QTL do not exist within the species, because the populations evaluated represent only a small proportion of the global diversity in maize [7]. Nevertheless, these results are consistent with QTL analysis of a large and genetically diverse mouse population, which also revealed a genetic architecture characterized by many genes of relatively small effects for many quantitative traits [8]. Similarly, in *Drosophila*, the relatively simple genetic architecture that appeared in initial genome scans of small populations has been revealed to be substantially more complex upon fine-mapping [1].

What causes the remaining genetic variation that is unexplained by QTL in large samples? One possibility is that there are many QTL with very small effects, as assumed in classical models of quantitative genetics, and these remain undetected even with very large sample sizes. Another possibility is that higher-order epistatic interactions, which

Figure 1



Genetic sampling and genetic heterogeneity complicate analyses of genetic architecture in plants. As an example, genetically diverse maize inbreds can be classified into genetically similar subgroups such as Temperate Stiff Stalk, Temperate Non-Stiff Stalk, and Tropical/Subtropical. QTL results from two independent mapping populations developed from the B73 $\times$ Mo17 cross can differ because of genotypic sampling. QTL results from unrelated populations, such as B73 $\times$ Mo17 and B104 $\times$ NC300, can differ because of both genetic heterogeneity and genotypic sampling.

are refractory to QTL mapping, control substantial amounts of genetic variation, although digenic (two-locus) epistasis was explicitly tested for and ruled out by Laurie *et al.* [2].

It is likely that many genes that have small effects will be missed in QTL mapping studies because researchers need to balance the chances of missing small effect QTL with false-positive QTL declarations. Recent advances in statistical methods to control the QTL false-discovery rate should provide a better balance between declaring too many false-positive QTL and sacrificing power to detect QTL that have smaller effects [9]. QTL mapping studies will be substantially improved when these methods are incorporated into model selection strategies and software, as model selection criteria in popular QTL software packages do not prevent fitting too many QTL into models [10].

Two complementary approaches are being pursued to improve our understanding of complex traits: first, detailed analysis of a few QTL via fine mapping and cloning, and second, large-scale diversity-based QTL analyses of multiple populations that have broader inference spaces.

Identifying DNA sequence variants underlying QTL

The ultimate objective of QTL mapping is to identify the causal genes, or even the causal sequence changes, the quantitative trait nucleotides (QTN). While this remains a major challenge, it has been achieved in a few instances [3,4]. Most QTN identification has proceeded from whole-genome scans to initially localize QTL to 10–20 cM intervals, followed by high-resolution genetic mapping, and finally, detailed genetic and complementation analyses of the sequence variants that co-segregate with the QTL [3,4]. Fine-mapping has been accomplished by selecting rare recombinants in the region of interest from very large populations that are nearly isogenic outside of the targeted region [4,11,12]. Identification of candidate genes and enrichment of markers within small targeted genomic regions should be greatly aided by growing genomics database resources that permit information sharing across populations and species thanks to genomic synteny. Association mapping provides an alternative method for resolving QTL down to specific DNA sequences, if linkage disequilibrium (LD) is low in the population/genome region studied, population structure is properly

accounted for, and good candidate genes can be selected for analysis [13].

QTL fine-mapping sometimes reveals surprises, such as the existence of distinct genes whose combined effects contribute to the QTL identified in whole-genome scans [1,11], distant upstream non-coding enhancer sequences that regulate phenotypic effects of a QTL [14^{••}], and substantive structural differences between the alternate alleles in the QTL region [15]. Identifying the genes or sequence variants that underlie QTL provides a powerful tool for investigating the contribution of specific genes to the overall genetic architecture of complex traits [12^{••}].

Diversity-based QTL mapping

QTL mapping in biparental populations reveals only a slice of the genetic architecture for a trait because only alleles that differ between the two parental lines will segregate (Figure 1). Therefore, more comprehensive analyses of genetic architecture require consideration of multiple populations that represent a larger sample of the standing genetic variation in the species.

One approach to enhancing the inference space of QTL studies is meta-analysis, whereby the results of multiple experiments are summarized and evaluated statistically [16,17]. These studies provide useful summaries of genetic architecture; however, they implicitly assume QTL additivity and are complicated by ascertainment bias and by differences in experimental techniques, statistical analyses, and environments among studies.

Another approach is to map QTL in extant breeding lines with known pedigree relationships. Given pedigree and marker information, the probability that different lines in complex populations share identical-by-descent QTL alleles can be defined, permitting estimation of the effects of each QTL allele [18]. Crepieux *et al.* [19[•]] used a similar method, but accounted for identity-by-descent between progeny of different crosses using random background markers.

Association mapping provides an alternate route to identifying QTL that have effects across a broader spectrum of germplasm, if false-positives that are caused by population structure can be minimized [13]. Whole-genome association scans are expected to be effective when LD and marker density are sufficiently great that random markers will have a good chance of being in disequilibrium with QTL across diverse genetic materials. This appears to be the case in diverse samples of *Arabidopsis* ecotypes [20]. Surprisingly, association mapping with a limited number of random markers identified QTL that have consistent effects across multiple heterotic groups in maize [21]. In this case, LD was probably extensive within heterotic groups, resulting in increased power to

detect QTL using linked markers, at a cost of lower resolution. Further, large-scale population structure effects were minimized by analyzing associations separately for each heterotic group, and finer-scale population structure was controlled by explicitly incorporating pedigree relationships between lines in the analysis [21]. This approach has been generalized by Yu *et al.* [13].

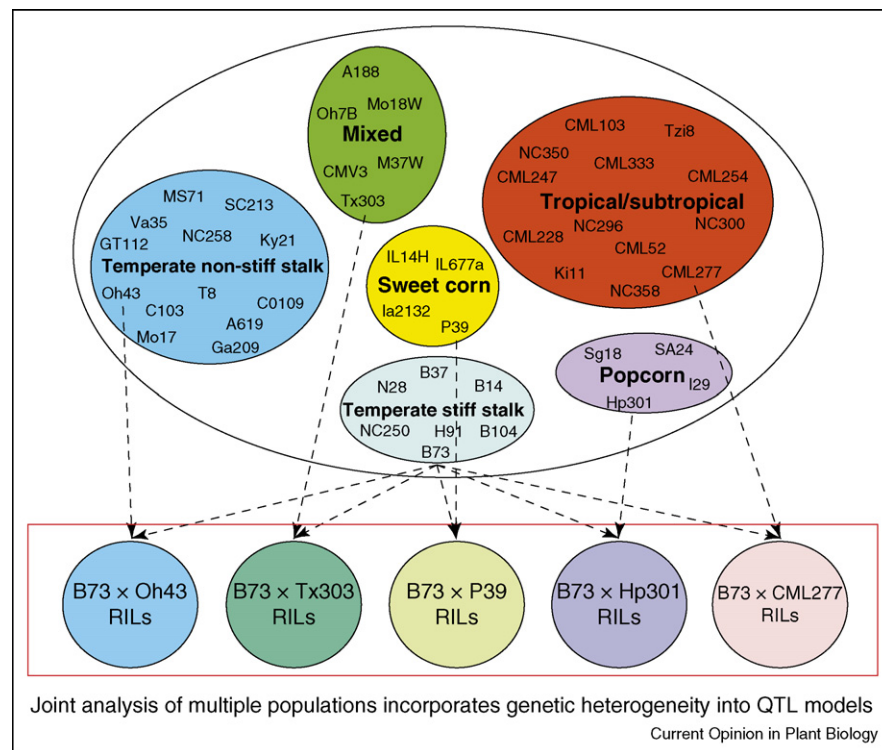
Finally, methods are being developed to jointly analyze QTL in multiple populations. Cross-population analyses have been made by comparing QTL maps from different populations studied in common environments [22], but more powerful analyses could be performed by taking into account the information shared across populations. This can be achieved by the use of interconnected populations that share parents [23^{••},24,25] or by defining regions that are identical by descent across populations based on haplotype sharing (Figure 2). Multi-population analyses can explicitly incorporate genetic heterogeneity into the QTL models, permitting the identification of QTL alleles that are carried by specific parental lines, the ranking of QTL allelic values across parents, and tests of epistatic interactions of QTL with the genetic background [23^{••}].

Epistasis

Gauging the importance of epistatic gene interactions is crucial because epistasis complicates the genotype–phenotype relationship, and competing models of evolution differ largely in the importance accorded epistasis [26]. Recent studies in *Arabidopsis* [26] and rice [27] suggest that epistatic QTL effects are more important than additive QTL for fitness traits. By contrast, however, studies designed to explicitly model epistatic interactions in maize revealed that epistasis was of little or only moderate importance for quantitative traits [6,23^{••},28]. These contrasting results might be due in part to the relative importance of epistatic effects in predominantly inbreeding and predominantly out-crossing species, and in part to differences in modeling procedures.

Incorporating epistasis exacerbates the statistical difficulties of QTL modeling. The number of possible marker pair interactions to be tested can greatly outnumber the genotypic sample size. One option is to test for epistatic effects only between QTL that have significant main effects, which greatly eases the model selection problems but has no biological justification. Another option is to test all possible pairs of loci for interactions and then use model selection techniques to fit significant epistatic pairs simultaneously with additive-acting QTL in a common model, with no guarantee that an optimal model is selected [28]. Recently, Xu [29[•]] addressed this dilemma with a Bayesian-based modeling approach that avoids model selection problems and fits all main effects and all interactions simultaneously, but shrinks the estimates of effects with little statistical support to zero.

Figure 2



Joint analysis of multiple related populations can incorporate genetic heterogeneity into the study of genetic architecture. As an example, 26 populations derived by crosses between the common parent B73 and 26 different inbred lines sampled to represent much of the diversity among the other subgroups of maize are being investigated (www.panzea.org). Five of the 26 populations are shown.

An alternative to whole-genome scans for epistasis is high-resolution genetic analysis of small genome regions. Kroyman and Mitchell-Olds [12^{••}] analyzed a 1-cM region in *Arabidopsis* that showed no effect on fitness in an initial genome scan. After high-resolution fine-mapping, they discovered two QTL for fitness in the region, each exhibiting strong epistasis with the genetic background. Depending on the background, allele effects at these QTL reverted from positive to negative. If this genome region is representative, we can expect substantial complexity of QTL and genetic architecture at the fine scale.

Heterosis

Heterosis, or hybrid vigor, is a key feature of the genetic architecture of fitness traits in predominantly outcrossing plants such as maize. Heterosis is of less importance in primarily self-fertilized plants, although it is exploited in tomato and rice breeding. Maize geneticists have long accepted that dominance of favorable alleles at complementary sets of loci in different lines is the basis for most heterosis in maize, although repulsion-phase disequilibrium between dominant alleles at linked loci results in the appearance of overdominance [30]. By contrast, recent QTL studies in rice [27] and tomato [31^{••}] suggest that overdominance is important for het-

eriosis in those species. Repulsion-phase linkage of dominant alleles at complementary loci cannot be ruled out in these studies, however, because the QTL regions defined still contain numerous genes whose combined dominant effects could appear as overdominance.

A new approach to studying hybrid vigor at the molecular scale is to survey gene expression as a phenotype in F₁ hybrids and their parental inbreds. Analysis of thousands of genes reveals that gene expression in a commercially important maize hybrid is mostly near the average level of the parental inbreds [32,33^{••}]. Where heterosis for expression was observed, it was mostly within the range of the parental inbreds, with less than 3% of differentially expressed genes exhibiting heterosis beyond the range of the parents.

Gene expression analyses will be an important tool for unraveling genetic architecture and the connections between genotypic and phenotypic variation, but the results of such studies require careful interpretation. The ratio of gene expression in F₁ progeny to the average of the inbred parents commonly has been interpreted as an estimate of the ratio of dominance to additive effects at a locus, and thus as a test for overdominance [32,33^{••}]. This approach is flawed, however, as gene expression

levels are phenotypes that can be affected by numerous loci beyond the specific gene whose mRNA level is being considered. Heterotic expression levels could be due to dominance at, or epistasis between, multiple loci, including *trans*-regulators. To directly relate gene action to expression phenotypes will require genetic mapping approaches, such as expression QTL mapping [34], although this will remain challenging because of the need to assay expression levels in large numbers of genotypic samples. Finally, studies of heterosis at the level of gene expression will illuminate only part of the black box that connects genotype and phenotype, as post-transcriptional regulation, metabolic interactions, and whole-plant physiological processes surely exert important effects on heterosis.

Conclusions

Unraveling the genetic architecture of complex traits in plants will require many more studies along the parallel tracks of detailed analysis of small genome regions and large-scale investigations of the genome function across diverse populations. Future research goals should also include modeling relationships between specific environmental factors and particular QTL [35], and modeling the impact of specific genome regions on overall genetic correlations between different traits [10]. Concurrent advances in experimental approaches, statistics, and cost-effective genomics tools will be required. Accurate phenotypic analyses will still be required to relate genotypic to phenotypic variation. Improved understanding of the evolution of natural plant populations and a positive impact on the breeding of domesticated crops should follow advances in the study of genetic architecture.

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